PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

ARENA

submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CONTENTS

S.No.	Title	Page No.
1.	Acknowledgement	4
2.	List of Figures and Tables	5
3.	Abstract	6-7
4.	Introduction	8
5.	Methodology	9
6.	Tools and Technologies	10
7.	Implementation	11
8.	Major Findings/Outcomes/Output/Results	12
9.	Conclusion and Future Scope	13
10.	References	14
11.	Appendices	15

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"ARENA"** in partial fulfillment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Mr Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 30/4/2026

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Finally, I express my gratitude to everyone, directly or indirectly, who contributed to the successful completion of this project. This work has been an important learning experience and a valuable step in my academic and technical journey.

List of Figures and Tables

Figures

- Figure 1: ARENA System Architecture
- Figure 2: Gameplay Loop Flowchart
- Figure 3: Enemy Spawn and Movement Logic
- Figure 4: Touch Control and Collision System
- Figure 5: Game Performance Optimization Model

Tables

- Table 1: Technology Stack Overview
- Table 2: Core Gameplay Mechanics
- Table 3: Performance Optimization Techniques
- Table 4: Expected Results and Benchmarks

Abstract

ARENA is a minimalist high-speed survival arcade game designed around pure mechanical skill, reaction speed, and visual rhythm. The project follows a “less is more” philosophy, focusing on elegant gameplay mechanics rather than complex progression systems or content-heavy structures. The central idea behind the game is to create an engaging survival experience where challenge emerges through escalating pressure, precision movement, and player mastery rather than artificial rewards or monetization-driven engagement loops.

Unlike conventional mobile games that often rely on pay-to-win systems, repetitive grinding, or overstimulating reward mechanics, ARENA emphasizes fairness, lightweight architecture, responsive interaction, and skill-based challenge. The gameplay experience is built to test reflexes and adaptability in real time while maintaining a visually clean and immersive aesthetic using high-contrast neon visuals and minimalist design principles.

The game features a glowing player-controlled sphere navigating a hostile environment where aggressive enemy nodes continuously spawn and pursue the player. As survival duration increases, the difficulty scales dynamically through increased enemy pressure, creating a progressively demanding arcade challenge. Players earn scores based on survival performance, encouraging replayability through mastery rather than forced retention systems.

Built using Expo SDK, React Native, and TypeScript, the project integrates optimized animation systems, efficient frame-synced game loop logic, touch-responsive control handling, and low CPU architecture to ensure smooth performance across a wide range of mobile devices. Performance optimization remains a core focus of the implementation, with architecture choices aimed at minimizing unnecessary processing while preserving gameplay responsiveness.

This report presents the problem context, system design methodology, technical implementation strategy, major findings, expected outcomes, and future scope of the project. It demonstrates how minimalist game design, combined with efficient engineering practices, can create a compelling and scalable mobile gaming experience.

Introduction

Modern mobile games often prioritize engagement loops, monetization systems, and retention-driven mechanics over mechanical purity, technical efficiency, and genuine player skill. Many contemporary titles depend heavily on complex progression models, excessive visual clutter, repetitive gameplay patterns, and resource-intensive systems that can compromise both performance and gameplay quality. In many cases, the focus shifts from creating meaningful player experiences to maximizing engagement metrics.

ARENA was conceived as an alternative to this approach—a minimalist survival arcade experience centered on reflex-driven gameplay, visual clarity, and optimized engineering. Rather than relying on content overload or reward-based compulsion, the game creates challenge through simple but escalating mechanics where survival depends on precision, awareness, and reaction speed.

The game challenges players to survive increasing enemy pressure while maintaining fast, responsive interaction in a visually clean and high-contrast environment. Its design emphasizes the idea that strong gameplay can emerge from focused mechanics supported by efficient technical architecture.

This project explores how elegant game design principles, combined with performance-oriented engineering, can produce a compelling mobile gaming experience that is both engaging and technically lightweight.

Methodology

The project follows a modular game development methodology consisting of:

1. Concept Design
Define gameplay objective and mechanics
Establish minimalist visual philosophy
Plan survival and scoring systems

2. System Architecture Design
Game loop structure
Enemy behavior logic
Collision detection planning

3. Development Phase
Core gameplay coding
Touch control integration
Animation and performance optimization
4. Testing and Evaluation
Performance benchmarking
Gameplay balancing
Device responsiveness testing
5. Refinement
Visual polish
Difficulty tuning
Stability improvements

Tools and Technologies

Core Framework

Expo SDK 54
React Native 0.81.5
TypeScript

Performance Engineering

React Native Animated API
useNativeDriver optimization
requestAnimationFrame game loop
useRef state handling
PanResponder touch system

Navigation and Infrastructure

Expo Router
EAS Build Services
Android Gradle Build System

Design Tools

Figma (UI references)
VS Code
Git and GitHub

Implementation

The ARENA system consists of the following modules:

1. Player Control Module
Touch-based direct movement
Latency-free motion tracking
Responsive interaction system

2. Enemy Spawn System
Boundary spawning logic
Dynamic spawn scaling
Homing movement behavior

3. Collision and Survival Engine
Collision detection processing
Survival condition evaluation
Real-time score updates

4. Game Loop Engine
requestAnimationFrame execution
Frame-synced updates
Performance-optimized calculations

5. Visual Rendering System
Neon glow effects
Micro-animations
Responsive layout rendering

Major Findings / Outcomes / Results

The project demonstrated:

Smooth gameplay can be achieved using lightweight mobile architecture.

High-frequency logic can operate efficiently without excessive React re-renders.

Native-driven animation significantly improves responsiveness.

Minimalist mechanics can produce engaging survival gameplay.

The system performs efficiently even on low-spec mobile hardware.

Expected output includes a playable optimized survival arcade prototype with strong responsiveness and scalable gameplay difficulty.

Conclusion and Future Scope

ARENA successfully presents a compact yet technically optimized survival arcade game that prioritizes performance, responsiveness, and pure skill-based gameplay.

The project proves that minimalist game systems can deliver both strong aesthetics and engaging mechanics without technical bloat.

Future Scope

Potential enhancements include:

Multiplayer survival modes

Procedural enemy behavior systems

Power-up mechanics

Leaderboards and ranked survival modes

AI-driven adaptive enemy patterns

Cross-platform deployment expansion

The project has potential for further commercial development and advanced mobile game research.

References

Sample references:

React Native Official Documentation

Expo SDK Documentation
TypeScript Handbook
Game Programming Patterns — Robert Nystrom
Mobile Performance Optimization Research Papers